

TRIE (YU YANG)

Undergraduate Researcher in Computer Science
University of British Columbia
Vancouver, BC

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Research Interests

Human-computer interaction, human-robot interaction, human-centered artificial intelligence, responsible artificial intelligence, interactive intelligent systems

Academic History

University of British Columbia, Vancouver, BC
Bachelor of Science, Computer Science

Expected 2027

Publications & Manuscripts

[1] **Yang, Y.**, Kim, I.-J., & Yoon, D. *PASTA: A Scalable Framework for Multi-Policy AI Compliance Evaluation*. Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26), 2026.

[2] **Yang, Y.**, Shen, A., Sayara, A., Kim, I.-J., & Yoon, D. *Negotiating Robot Autonomy to Preserve User Responsiveness: A User Enactment Study of Personal Security Robots for Women's Safety*. Manuscript submitted to ACM CHI 2027, under review.

Honors & Awards

2026 CRA Outstanding Undergraduate Researcher Award, Honorable Mention, Computing Research Association

2026 Work Learn International Undergraduate Research Award (WLIURA), University of British Columbia

2025 Work Learn International Undergraduate Research Award (WLIURA), University of British Columbia

Academic Service & Activities

2027 Invited reviewer, ACM CHI 2027

Full paper presenter, ACM CHI 2026, Barcelona

Invited guest speaker, Amazon AWS RAI Science Seminar

Invited guest speaker, Salish SIGCHI Celebration of Research, Vancouver

Research Experience

Socius Lab, University of British Columbia, Vancouver, BC

2025–present

Undergraduate Research Assistant

Supervisor: Dongwook Yoon

Developed an LLM-driven compliance checker for AI systems; conducted an expert evaluation and a 12-participant mixed-methods study. First-author ACM CHI 2026 full paper.

Investigating design of personal-security robots for women's safety using user-enactment methods with 25 participants, with a focus on user agency vs. robot autonomy. First-author ACM CHI 2027 full paper submission under review.

Contribute across research design, system development, study execution, qualitative and mixed-methods analysis, and academic writing.

Department of Computer Science, University of British Columbia, Vancouver, BC

2026–present

Undergraduate Research Assistant

Supervisor: Caroline Lemieux

Developing a TypeScript-focused subtool for DocPrism, an LLM-supported system for analyzing code-documentation quality and consistency.

Conducting qualitative analysis to evaluate tool quality and identify limitations in generated assessments.

Professional Experience

Optum, UnitedHealth Group, Vancouver, BC

2024–2025

Software Engineering Intern

Led GitHub and CI/CD migration work for the SI Cloud Archive codebase in an Agile environment.

Contributed to engineering and automated testing for a **C# DICOM** library handling streaming medical-data metadata for GCP integration.

Refactored and restored a C#/SQL Traceability Matrix tool for internal operational use.

Tang Diabetes Lab, UBC Faculty of Medicine, Vancouver, BC

2024–2025

Digital Platform Assistant

Supported onboarding of Type 1 diabetes patients to the REACHOUT app and provided technical guidance.

Conducted data analysis in Jupyter Notebook to inform app improvements.

Robotics Experience

Ten Ton Robotics, FIRST Robotics Competition Team 6486, West Vancouver

2020–2021

Developed human-operated robot functionality using WPILib and Java while leading a team of 10+ students; team qualified for FRC Provincials 2021.

Designed and implemented autonomous shooting features for the 2021 FIRST BC provincial competition.

Technical Skills

Programming: Python, C++, C#, Java, JavaScript/TypeScript, SQL, Swift, HTML/CSS

Research / Data: Pandas, NumPy, Jupyter, qualitative/quantitative analysis, mixed-methods research

Tools / Platforms: Git, Figma, Azure, GCP, MySQL/SQLite/MariaDB/MongoDB